

Problem 1. Let $\vec{a} = [3 \ 4]$ and $\vec{b} = [1 \ -2]$

Write both vectors in column vector notation and find their magnitudes

Solution.

$$\vec{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\vec{b} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$|\vec{a}| = \sqrt{3^2 + 4^2} = 5$$

$$|\vec{b}| = \sqrt{1^2 + (-2)^2} = 2.23$$

□

Problem 2. Let $\vec{a} = [5 \ 0 \ -6 \ 8]$ and $\vec{b} = [11 \ -12 \ -4 \ 7]$

Write both vectors in column vector notation and find their magnitudes.

Solution.

$$\vec{a} = \begin{pmatrix} 5 \\ 0 \\ -6 \\ 8 \end{pmatrix}$$

$$\vec{b} = \begin{pmatrix} 11 \\ -12 \\ -4 \\ 7 \end{pmatrix}$$

$$|\vec{a}| = \sqrt{5^2 + 0^2 + (-6)^2 + 8^2} = 11.2$$

$$|\vec{b}| = \sqrt{11^2 + (-12)^2 + (-4)^2 + 7^2} = 18.2$$

Show that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

$$\begin{aligned} \vec{a} \cdot \vec{b} &= (5 \ 0 \ -6 \ 8) \begin{pmatrix} 11 \\ -12 \\ -4 \\ 7 \end{pmatrix} = 5 \times 11 + 0 \times (-12) + (-6) \times (-4) + 8 \times 7 \\ &= 135 \end{aligned}$$

$$\begin{aligned} \vec{b} \cdot \vec{a} &= (11 \ -12 \ -4 \ 7) \begin{pmatrix} 5 \\ 0 \\ -6 \\ 8 \end{pmatrix} = 11 \times 5 + (-12) \times 0 + (-4) \times (-6) + 7 \times 8 \\ &= 135 \\ \vec{a} \cdot \vec{b} &= \vec{b} \cdot \vec{a} \end{aligned}$$

□

Problem 3. Let $\vec{u} = [2 \quad -1 \quad 4]$ and $\vec{v} = [-3 \quad 0 \quad 5]$
Find

1. $\vec{u} + \vec{v}$

Solution. $\vec{u} + \vec{v} = [2 \quad -1 \quad 4] + [-3 \quad 0 \quad 5] = [-2 \quad -1 \quad 9] \quad \square$

2. $2\vec{u} - \vec{v}$

Solution.

$$\begin{aligned} 2\vec{u} - \vec{v} &= 2[2 \quad -1 \quad 4] - [-3 \quad 0 \quad 5] \\ &= [4 \quad -2 \quad 8] - [-3 \quad 0 \quad 5] \\ &= [7 \quad -2 \quad 3] \end{aligned}$$

$\square$

Problem 4. Let $\vec{x} = [8 \quad -4 \quad 2 \quad 5]$, $\vec{y} = [-4 \quad 4 \quad 0 \quad 7]$ and $\vec{z} = [2 \quad 3 \quad -2 \quad 1]$
Find

$$\vec{x} + 3\vec{y} - 2\vec{z}$$

Solution.

$$\begin{aligned} \vec{x} + 3\vec{y} - 2\vec{z} &= [8 \quad -4 \quad 2 \quad 5] + 3[-4 \quad 4 \quad 0 \quad 7] - 2[2 \quad 3 \quad -2 \quad 1] \\ &= [8 \quad -4 \quad 2 \quad 5] + [-12 \quad 12 \quad 0 \quad 21] - [4 \quad 6 \quad -4 \quad 2] \\ &= [-8 \quad 10 \quad 24] \end{aligned}$$

$\square$

Problem 5. Let $\vec{a} = [1 \quad 2]$ and $\vec{b} = [2 \quad 3]$

Find the angle θ between them using the dot product *Solution.*

$$\begin{aligned} \cos\theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \\ &= \frac{1 \times 2 + 2 \times 3}{\sqrt{1^2 + 2^2} \sqrt{2^2 + 3^2}} = \frac{8}{8.06} = 0.99 \\ \theta &= \cos^{-1}(0.99) \approx 7^\circ \end{aligned}$$

$\square$

Problem 6. Given vector $\vec{v} = [6 \quad -8]$, find a unit vector in the same direction. *Solution.*

$$\begin{aligned} \text{unit vector } \vec{i} &= \frac{\vec{v}}{|\vec{v}|} \\ \vec{i} &= \frac{[6 \quad -8]}{\sqrt{6^2 + (-8)^2}} \\ &= \frac{[6 \quad -8]}{10} = [0.6 \quad -0.8] \end{aligned}$$

$\square$

Problem 7. Let $\vec{a} = [3 \quad 4]$ and $\vec{b} = [1 \quad 0]$. Find the projection of $\vec{a}$ onto $\vec{b}$. *Solution.*

$$\begin{aligned} \text{Proj.of } \vec{a} \text{ onto } \vec{b} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b} \\ &= \frac{3}{1} [1 \quad 0] = [3 \quad 0] \end{aligned}$$

□ Find the projection of $\vec{b}$ onto $\vec{a}$ Solution.

$$\begin{aligned}\text{Proj.of } \vec{b} \text{ onto } \vec{a} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a} \\ &= \frac{3}{25} [3 \ 4] = [9/25 \ 4/25]\end{aligned}$$

□

Problem 8. Let $\vec{p} = [2 \ -1, \ 3]$ and $\vec{q} = [0 \ 4, \ -1]$. Verify

$$\vec{p} \cdot \vec{q} = \vec{q} \cdot \vec{p}$$

Solution.

$$\begin{aligned}\vec{p} \cdot \vec{q} &= 2 \times 0 + (-1) \times 4 + 3 \times (-1) = -7 \\ \vec{q} \cdot \vec{p} &= 0 \times 2 + 4 \times (-1) + (-1) \times 3 = -7\end{aligned}$$

□

$$\vec{p} \cdot (\vec{p} + \vec{q}) = \vec{p} \cdot \vec{p} + \vec{p} \cdot \vec{q}$$

Solution.

$$\begin{aligned}\vec{p} \cdot (\vec{p} + \vec{q}) &= [2 \ -1 \ 3] \cdot ([2 \ -1 \ 3] + [0 \ 4, \ -1]) \\ &= [2 \ -1 \ 3] \cdot [2 \ 3 \ 2] \\ &= 4 - 3 + 6 = 7 \\ \vec{p} \cdot \vec{p} + \vec{p} \cdot \vec{q} &= [2 \ -1 \ 3] \cdot [2 \ -1 \ 3] + [2 \ -1 \ 3] \cdot [0 \ 4 \ -1] \\ &= (4 + 1 + 9) + (0 - 4 - 3) = 7\end{aligned}$$

□

Problem 9. Determine whether the vectors $\vec{a} = [1 \ 2]$, $\vec{b} = [2 \ 4]$ and $\vec{c} = [-1 \ -2]$ are linearly independent.

Solution. Check if any vector is a linear combination of the others:
Clearly, $\vec{b} = 2 \times \vec{a}$, and $\vec{c} = -1 \times \vec{a}$.

All three lie on the same line $\rightarrow$ Linearly dependent □

Problem 10. Are the vectors $\vec{u} = [3 \ 6]$ and $\vec{v} = [-2 \ -4]$ collinear?

Solution. Check if one is a scalar multiple of the other:
 $\vec{v} = (-2/3) \times \vec{u} \rightarrow$ Yes, $\vec{u}$ and $\vec{v}$ are collinear. □

Problem 11. Bonus Problem I

Let $\vec{a} = [1 \ 0 \ 0]$ and $\vec{b} = [0 \ 1 \ 0]$.

What is the angle between $\vec{a}$ and $\vec{b}$?

Solution.

$$\begin{aligned}\cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = 0 \\ \theta &= 90^\circ\end{aligned}$$

□ Find the vector cross product $\vec{a} \times \vec{b}$. Show that the resultant vector is in a direction normal to both $\vec{a}$ and $\vec{b}$ Solution.

Consider $\vec{i}, \vec{j}$ and $\vec{k}$ to be unit vectors along the respective directions

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \vec{i}(0 - 0) - \vec{j}(0 - 0) + \vec{k}(1 - 0) = \vec{k}$$

□

Problem 12. *Bonus Problem II*

Let $\vec{a} = [4 \ 1]$ and $\vec{b} = [2 \ 3]$. Find the area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$. *Solution.*

$$\begin{aligned} \text{Area} = |\vec{a} \times \vec{b}| &= \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} \\ &= 4 \times 3 - 1 \times 2 = 10 \end{aligned}$$

□